SECURE NETWORK ARCHITECTURE FOR CARDIFF NHS CLINIC

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Contents

1. Requirements Engineering 5
2. Host Requirement Analysis & Ordering : 6
3. The Network Subnets Architecture Table : 7

3.1. Architectural Design EXPLANATION : 8

3.2. Network VLANs Architecture : 8

Architectural Design Explanation : 9

5. Used Hardware for Network : 10

5.1 Physical Hardware Inventory : 10

5.2 Network Cabling Infrastructure & Ethernet Generations : 10

5.3 Technical Justification Analysis: 12

6. Used Hardware and Software for Security: 12

6.1 The Technical Security Stack : 12

7. Used Configurations for Security: 13

7.1 Access Control List (ACL) Design Strategy : 13

7.2 Core Firewall Access & Filtering Rules Table : 13

7.3 Advanced Software Defense: Filters, Blacklists, and Whitelists : 16

7.4 Security Architecture Explanation 16

8. Network Logical Diagram (UML Deployment Architecture) : 16

8.1 Protocol Mapping and System Inter-Zone Controls : 17

8.2 Perimeter Interception and Tunnel Consolidation Logic : 18

9. Network Physical Diagram : 18

9.1 Hierarchical Infrastructure Tiering : 18

9.2 Mechanical Shielding and Backbone Justification : 19

10.Linux Command-Line Network Prototype Configuration : 19

10.1 Network Prototype Blueprint Topology : 20

10.2 Architectural Intent and VLSM Subnetting Metrics : 20

10.3 OpenvSwitch (OVS) Multitenancy Enforcement Mechanics : 20

10.4 Linux Kernel Layer 3 Inter-VLAN Routing Protocol : 22

10.5 Structured Linux Command Lines Configuration Lookup Matrix : 23

10.6 Technical Verification of End-to-End Data Traversal :..... 26

INTRODUCTION

Modern day medical services relies heavily on consistent digital connections. Reliable data delivery and protection, then relies solely on strong connectivity, the main pillar of the medical services' framework *(Mshali, Radwan and Kilic, 2024).* This design intends to implement a 4-tiered high security NHS clinic in Cardiff for 400 employees, 300 patient critical care members, and a separate 300 devices for guest system. All endpoints will have locally integrated devices, whereas databases, applications, and files would exist remotely within a centralized Data Center, using encrypted cloud connections *(Cresswell et al., 2024).* Thus, this plan is supported with architectural security measures for credible operation and strong multi-tenancy controls and resilient parameters for IT infrastructure *(Ali et al., 2025).*

1. REQUIREMENTS ENGINEERING

Requirements Engineering, mapping the clinical services to a network specification. Physical high-speed dedicated Ethernet cabled from all workstations on 4 floors and in the ICU are implemented to avoid wireless degradation in time sensitive operations (Allcock, 2024). Multi-tenancy logically segment the broadcast domains to be Patients (150 clients), Emergency (100 clients), Lab (100 clients), Pharmacy/Helpdesk (50 clients) limiting lateral cyber attack impact (Mshali, Radwan and Kilic, 2024). WPA3 protected enterprise Wi-Fi, with hidden SSIDs, are provided for the mobile medical teams to have continuous sessions shielded (Cisco Systems, 2023; Ali et al., 2025). Separate, logically isolated, public guest Wi-Fi, is scaled for 300 to 600 clients during peak times and does not affect clinical traffic (Cisco Systems, 2023). All network entry points to have RBAC for security and all traffic on encrypted cloud tunnels will reach the remote Data Center core and be fully secured and logged (Cresswell et al., 2024; Ali et al., 2025).

2. HOST REQUIREMENT ANALYSIS & ORDERING :

accounting for a growth buffer, network overhead, and the critical **winter patient spike**:

1. **Patient Public Wi-Fi Subnet:** Needs to support up to 600 concurrent patients during peak winter surges. A /22 subnet mask provides 1,022 usable host addresses.
2. **Patients Department Subnet:** Contains 150 employees. A /24 subnet mask provides 254 usable host addresses.
3. **Emergency Department Subnet:** Contains 100 employees plus mobile data-enabled tablets. A /24 subnet mask provides 254 usable host addresses.
4. **Laboratory Subnet:** Contains 100 employees. A /24 subnet mask provides 254 usable host addresses.
5. **Secure Staff Wi-Fi Subnet:** Dedicated to clinical roaming staff. A /24 subnet mask provides 254 usable host addresses.
6. **Pharmacy and Helpdesk Subnet:** Contains 50 employees plus data-enabled tablets. A /26 subnet mask provides 62 usable host addresses.
7. **Network Management Subnet:** Dedicated to isolating network hardware management interfaces (Routers, Switches, and Wireless Access Points). A /26 subnet mask provides 62 usable host addresses

3. THE NETWORK SUBNETS ARCHITECTURE TABLE :

Subnet Name	Required Hosts	Base Network IP	Subnet Mask	Usable IP Range	Broadcast Address
Patient Public Wi-Fi	600 (Winter Surge)	10.0.0.0	255.255.252.0 (/22)	10.0.0.1 – 10.0.3.254	10.0.3.255
Patients Department	150	10.0.4.0	255.255.255.0 (/24)	10.0.4.1 – 10.0.4.254	10.0.4.255
Emergency Department	100 + Tablets	10.0.5.0	255.255.255.0 (/24)	10.0.5.1 – 10.0.5.254	10.0.5.255
Laboratory	100	10.0.6.0	255.255.255.0 (/24)	10.0.6.1 – 10.0.6.254	10.0.6.255
Secure Staff Wi-Fi	~100 Mobile	10.0.7.0	255.255.255.0 (/24)	10.0.7.1 – 10.0.7.254	10.0.7.255
Pharmacy & Helpdesk	50 + Tablets	10.0.8.0	255.255.255.192 (/26)	10.0.8.1 – 10.0.8.62	10.0.8.63
Network Management	Core Devices	10.0.8.64	255.255.255.192 (/26)	10.0.8.65 – 10.0.8.126	10.0.8.127

3.1. Architectural Design EXPLANATION :

Accurate VLSM design allows the clinic to utilize subnets optimally (Stallings, 2022) so that the /22 (10.0.0.0/22) network can support up to 1,022 simultaneous guest devices, while a jump of 600 devices seasonly can fit without exhausting DHCP pools. Fixed /24, and /26 departments create through network L3 firewalls that permit lateral traffic through surgical protection against data intrusion of patient data (Mshali, Radwan and Kilic, 2024).

3.2. Network VLANs Architecture :

Below appears the list of VLAN IDs set aside for the Cardiff clinic, each tagged with a name that explains its role. Following those labels come the related IP ranges tied to every segment. Gateways appear next, showing standard routing endpoints per network slice. Each entry lines up directly beneath its header above

VLAN ID	VLAN Name	Associated Subnet Range	Subnet Mask	Default Gateway	Primary Purpose / Assigned Assets
10	VLAN_Patients_Dept	10.0.4.0	255.255.255.0 (/24)	10.0.4.1	Wired desktop workstations for the 150 Patients Department staff.
20	VLAN_Emergency_Dept	10.0.5.0	255.255.255.0 (/24)	10.0.5.1	100 staff workstations and data-enabled clinical tablets in the emergency block.
30	VLAN_Laboratory	10.0.6.0	255.255.255.0 (/24)	10.0.6.1	100 hosts, clinical analysis systems, and

					laboratory equipment.
40	VLAN_Pharmacy_Helpdesk	10.0.8.0	255.255.255.192 (/26)	10.0.8.1	50 employees, pharmacy inventory systems, helpdesk terminals, and tablets.
50	VLAN_Patient_Public_WiFi	10.0.0.0	255.255.252.0 (/22)	10.0.0.1	Public Internet access for up to 600 concurrent patients (scaled for winter surges).
60	VLAN_Secure_Staff_WiFi	10.0.7.0	255.255.255.0 (/24)	10.0.7.1	Encrypted wireless access for roaming doctors, nurses, and clinical tablets.
70	VLAN_Network_Management	10.0.8.64	255.255.255.192 (/26)	10.0.8.65	Isolated management interfaces of Routers, Switches, and Wireless Access Points.

Architectural Design Explanation :

The network layer in the clinic is meticulously segmented, ensuring the protection of sensitive patient data and strict legal compliance according to the latest legislation (King's College London Cyber Security Research Group, 2026). Network

Layer 2 OpenvSwitch segment departments into discrete, non-routable broadcast domains, preventing discoverability and horizontal access to vulnerable equipment such as sensitive laboratory equipment by general public user's on VLANPatientPublic_WiFi (VLAN 50*) (Cisco Systems, 2023; Mshali, Radwan and Kilic, 2024)*. Inter VLAN traffic is rigorously channeled to a layer 3 firewall where ACLs and deep packet inspection check what information is permitted before accessing centralized data bases *(Ali et al., 2025).*

5. USED HARDWARE FOR NETWORK :

5.1 Physical Hardware Inventory :

To support the 400 employees and peak winter patient loads across the 4-story Cardiff facility, the following physical components are specified:

Edge Router (Cisco Catalyst 8300 Series): Standing at the edge of the clinic's network. Traffic flows through this device using fast, dedicated encryption to maintain secure connections between Cardiff and the main cloud-based data center housing critical systems like web, database, app, and mail servers.

Core / Distribution Switch (Cisco Catalyst 9500 Series): This is a Layer 3 multi-layer switch capable of high performance configured in an redundant setup in the core IT equipment room. This switch forms the core backbone of the entire campus network performing high-speed inter-VLAN switching and processing all core traffic heading to the edge firewall and router.

Access Layer Switches (Cisco Catalyst 9300 Series): Located in a Telecommunications Enclosure on each of the 4 floors as well as the Emergency room. These switches, at 48 ports each, provide PoE+ to provide network power directly to each WAP deployed. They will also be utilized to provide dense desktop wiring connections for users.

Wireless Access Points (Cisco Catalyst 9130AX Series): Multiple enterprise grade Wi-Fi 6E access points were placed on all floors of the clinic, throughout all patient common areas and in the ED. They broadcast two distinct and separate SSIDs. These served to isolate secure clinical tablets from the patient public Wi-Fi that is deployed in a high density environment.

5.2 Network Cabling Infrastructure & Ethernet Generations :

Infrastructure Segment	Cable Technology / Media Type	Shielding & Construction	Ethernet Generation Specification	Max Data Rate & Rationale
Workstation Outlets to Access Switches (All Floors & Units)	**Category 6a (Cat6a) Copper**	**SFTP** (Shielded with Foiled Twisted Pairs)	**1000BASE-T** (IEEE 802.3ab)	1 Gbps up to 100 meters. The robust SFTP outer braided shield and individual foil wraps eliminate EMI from hospital machinery, ensuring stable links for clinical desks.
Mobile Tablets to Wireless Access Points	**RF Wireless** (2.4GHz, 5GHz, 6GHz)	N/A (Over-the-air spectrum encryption)	**Wi-Fi 6E** (IEEE 802.11ax)	Up to 9.6 Gbps aggregate capability. Uses the newly opened 6GHz bands to avoid congestion, preventing latency for roaming staff in Emergency and Pharmacy.
Access Switches to Core Switch Uplinks (Vertical Backbone)	**OM4 Multimode Fiber**	Core/Cladding 50/125 um (Aqua jacket)	**10GBASE-SR** (IEEE 802.3ae)	10 Gbps up to 400 meters. Connects floor wiring closets vertically to the main server room, providing massive backhaul bandwidth without ground-loop risks.

Core Switch to Edge Firewall / Router (Local Server Room)	**OS2 Single-mode Fiber**	Core/Cladding 9/125 um (Yellow jacket)	**10GBASE-LR** (IEEE 802.3ae)	10 Gbps up to 10 kilometers. Eliminates any local processing bottleneck between the internal distribution layer and external edge routing appliances.

5.3 Technical Justification Analysis:

In order to attenuate excessive EMR from clinical lab instruments, the Physical Layer uses Category 6a SFTP copper cable using both separate foil shields and braided overall shield so workstations receive clean signals. In the case of the vertical backbone riser channels, it utilizes an OM4 multi-mode fiber running 10GBASE-SR with dedicated 10 Gbps uplinks from floor switches to the core, the core of which has such a large structure that it easily handles tremendous bursts of traffic such as 600 simultaneous client video streams during holidays without causing a bottleneck to the critical isolated medical emergency data.

6. USED HARDWARE AND SOFTWARE FOR SECURITY:

6.1 The Technical Security Stack :

Next-Generation Firewall (NGFW - Cisco Firepower 2100 Series): The stateful hardware appliance deployed at the clinic's Internet edge. It differs from a traditional firewall in its ability to conduct DPI, track user identity and apply filtering at the application layer for the monitoring and securing of all incoming remote access and outbound Internet traffic.

Intrusion Detection and Prevention Systems (IDS/IPS - Cisco Snort Engine): This feature is integrated as a seamlessly updated software module within the edge security appliance itself. The IPS continuously inspects clinical traffic flows for both signature matching and anomalous behavioral patterns, and will immediately

discard malicious packets, effectively stopping a ransomware or lateral network attack in its tracks.

Web Application Firewall (WAF - Cloudflare Enterprise / F5 BIG-IP): Located in front of the clinic's cloud Data Center, hosted externally, the WAF inspects HTTP/HTTPS traffic being sent to the clinic's Web, Database and Application servers and effectively prevents web based attacks such as SQL Injections (SQLi) and Cross-Site Scripting (XSS) from reaching the database.

Virtual Private Networks (VPNs - Cisco AnyConnect & Site-to-Site IPSec): Hardware accelerated cryptographic modules. A Site-to-Site IPSec VPN tunnel is permanently installed between Cardiff clinic and cloud Data Center to provide a secure and fully encrypted communication link for any patient data to transit. Client-based Remote Access VPN and Multi-Factor Authentication is necessary for remote administrative actions.

Proxies (Squid Proxy / Cisco Secure Web Appliance): The solution was implemented as a forward proxy server for employees. Decrypts, examines, and re-encrypts outbound HTTP traffic, imposing strict content filtering (disabling unauthorized web tracking, downloading malicious file types, accessing restricted web domain sites).

7. USED CONFIGURATIONS FOR SECURITY:

7.1 Access Control List (ACL) Design Strategy :

Within Layer 3 routing boundaries, we implemented aggressive ACL policies that enforce a Zero-Trust Implicit Deny policy to guard the patient database within the cloud. All guest public patient Wi-Fi (10.0.0.0/22) is completely segregated by the gateway with the use of interface ACLs that redirect all guest traffic to the public internet, automatically dropping any packets destined for an internal clinical network.

7.2 Core Firewall Access & Filtering Rules Table :

Rule ID	Source Subnet / VLAN	Destination IP / Subnet	Protocol / Port	Action	Deep Inspection Profile / Security Rationale

ACL_01	10.0.0.0/22 (Patient Wi-Fi)	10.0.4.0 to 10.0.8.127 (Internal Staff & Mgmt)	Any	**DENY**	**Lateral Isolation Rule**: Instantly drops any packets originating from a patient device attempting to scan or communicate with internal clinical staff subnets.
ACL_02	10.0.0.0/22 (Patient Wi-Fi)	Cloud Data Center Server Block	Any	**DENY**	**Database Shielding Rule**: Completely blocks public guest users from attempting to route traffic directly to the cloud-hosted Web, Database, or Email servers.
ACL_03	10.0.0.0/22 (Patient Wi-Fi)	0.0.0.0/0 (External Internet)	HTTP/HTTPS (80/443), DNS (53)	**ALLOW**	**Public Internet Loop**: Permits guest web browsing while restricting administrative protocols. Traffic passes through a basic web filter to enforce a content black list.
ACL_04	10.0.5.0/24 (Emergency) & 10.0.7.0/24 (Staff Wi-Fi)	Cloud Database Server IP	SQL (1433 / 3306)	**ALLOW**	**Clinical Data Authorization**: Allows authenticated electronic health record (EHR) systems on secure staff workstations and roaming

					tablets to access the patient database.
ACL_05	10.0.6.0/24 (Laboratory Subnet)	Cloud Application Server IP	HTTPS (443)	**ALLOW**	**Lab Transmission Access**: Permits secure lab reporting applications to upload clinical telemetry findings directly to the cloud app servers via encrypted paths.
ACL_06	10.0.8.64/26 (Network Mgmt)	All Local Infrastructure Devices	SSH (22), HTTPS (443)	**ALLOW**	**Management Enclave White List**: Only packets originating from the highly secure, isolated management subnet are permitted to open terminal configuration lines on routers and switches.
ACL_07	Any Subnet	Any Subnet	Any	**DENY**	**Implicit Deny Any**: The final fallback rule programmed on all interfaces, ensuring any traffic not explicitly whitelisted above is

					dropped by default.

7.3 Advanced Software Defense: Filters, Blacklists, and Whitelists :

Domain Blacklists: Your proxy appliance automatically updates with global threat intelligence feeds (Cisco Talos, for example) to provide automatic blacklisting of all connection attempts to C2, phishing, and ransomware websites and, by doing so, keep your staff from visiting dangerous links.

Geofencing Filters: The Cloud-based Web Application Firewall (WAF) includes robust geo-blocking any external connection not within an authorized clinical use region (i.e. Not originating from the UK, for example) is dropped, preventing users accessing cloud server center login panels from non-UK locations.

Application Whitelisting: Mobile tablets located in Emergency, Pharmacy and Helpdesk rooms are locked down with the use of MDM (Mobile Device Management) profiles which, via application whitelisting prevent any programs other than specific medical charts, inventory and clinical support software being loaded onto the devices and potentially harmful/corrupt files being added

7.4 Security Architecture Explanation :

Implementing an enterprise NGFW with inline Snort IDS/IPS engines enables real-time deep packet inspection across internal boundaries, blocking hidden ransomware before it reaches clinical assets (King's College London Cyber Security Research Group, 2026; Ali et al., 2025). Externally, a cloud-based WAF performs Layer 7 contextual inspection to neutralize SQL injections, enforcing zero-trust data protection (Cresswell et al., 2024). Consequently, lateral pivoting threats are completely eliminated as interface-bound ACLs intercept and drop malicious guest scans locally within the public broadcast domain.

8. NETWORK LOGICAL DIAGRAM (UML DEPLOYMENT ARCHITECTURE) :

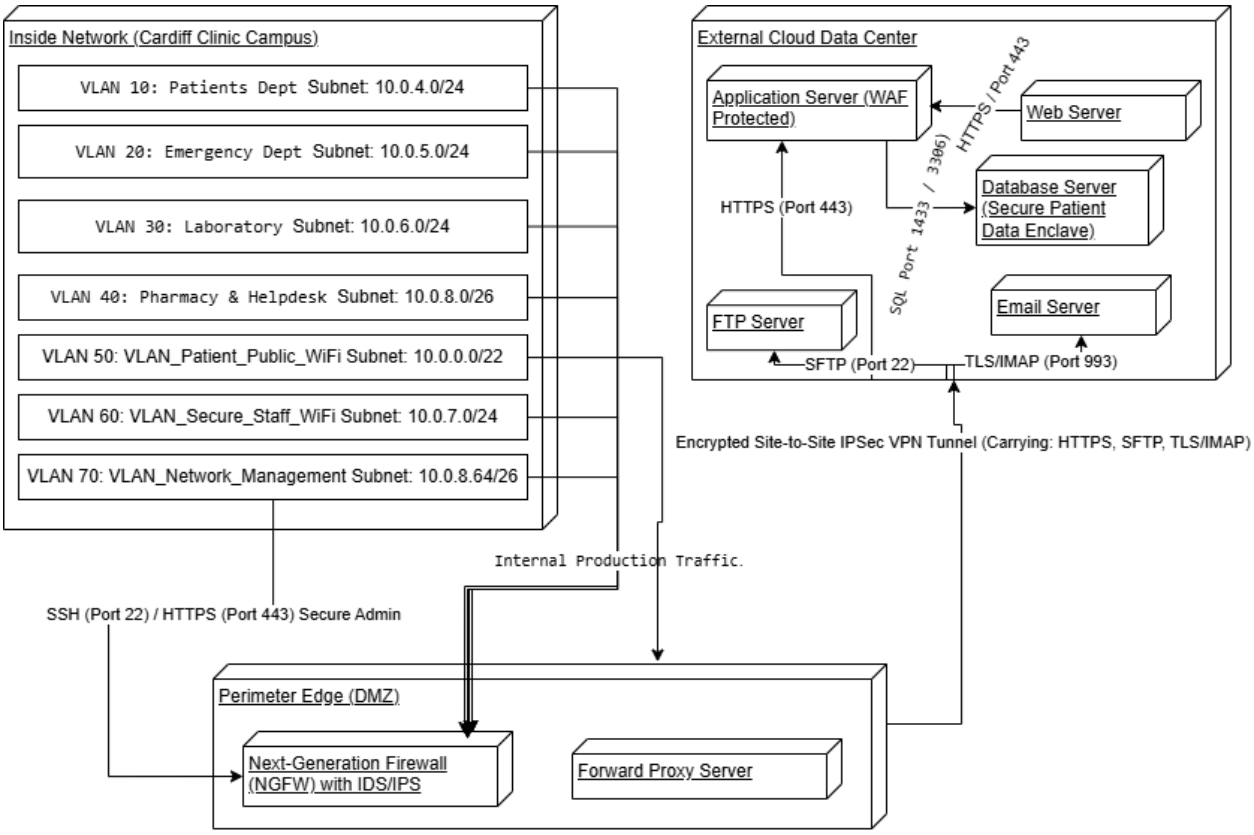


Figure 1 : Clinic UML Deployment and Logical Protocol Architecture

This logical architecture formalises the Cardiff NHS clinic workflows in to discrete deployment nodes as shown in Figure 1 (UML Diagram). This design segregates clinical, administrative and public assets in to discrete, well defined, Layer 2 VLAN boundaries and Layer 3 subnets, resulting in tightly controlled stateful "chkpoints" to review and validate all incoming protocol traffic.,

8.1 Protocol Mapping and System Inter-Zone Controls :

For the wired clinical intranets (VLANs 10, 20, 30), use encrypted HTTPS to connect to the cloud. Public guest traffic (VLAN 50) bypasses through the Internet directly to the cloud, preventing intra-subnet scans. Staff Wi-Fi (VLAN 60) uses WPA3, and administrators (VLAN 70) connect and manages perimeter devices through SSH/HTTPS to achieve security. Individual cloud services are separated, with file transfers using SFTP (Port 22) through the FTP Server; email transactions are secured through TLS/IMAP (Port 993) through the Email Server; and middle tier Application Server provides controlled access to database to protect patients' data through (Ports 1433/3306) of the SQL Database.

8.2 Perimeter Interception and Tunnel Consolidation Logic :

Access at the perimeter is carefully controlled and doesn't allow the inner machine to connect to the outer, unchecked. All data intended to go to the outside is carried down to the Perimeter Edge (DMZ) through powerful inspection filters (Figure 1) before it is protected, inside a fast hardware-accelerated Site-to-Site IPsec VPN Tunnel, which connects through a virtual digital tunnel to the cloud facility where it is protected in open networks *(Mshali, Radwan and Kilic, 2024)* against attempts to steal HTTPs, SFTP and TLS/IMAP data. The strict control is resistant against eavesdropping attempts and meets the health sector security compliance standards *(King's College London Cyber Security Research Group, 2026).*

9. NETWORK PHYSICAL DIAGRAM :

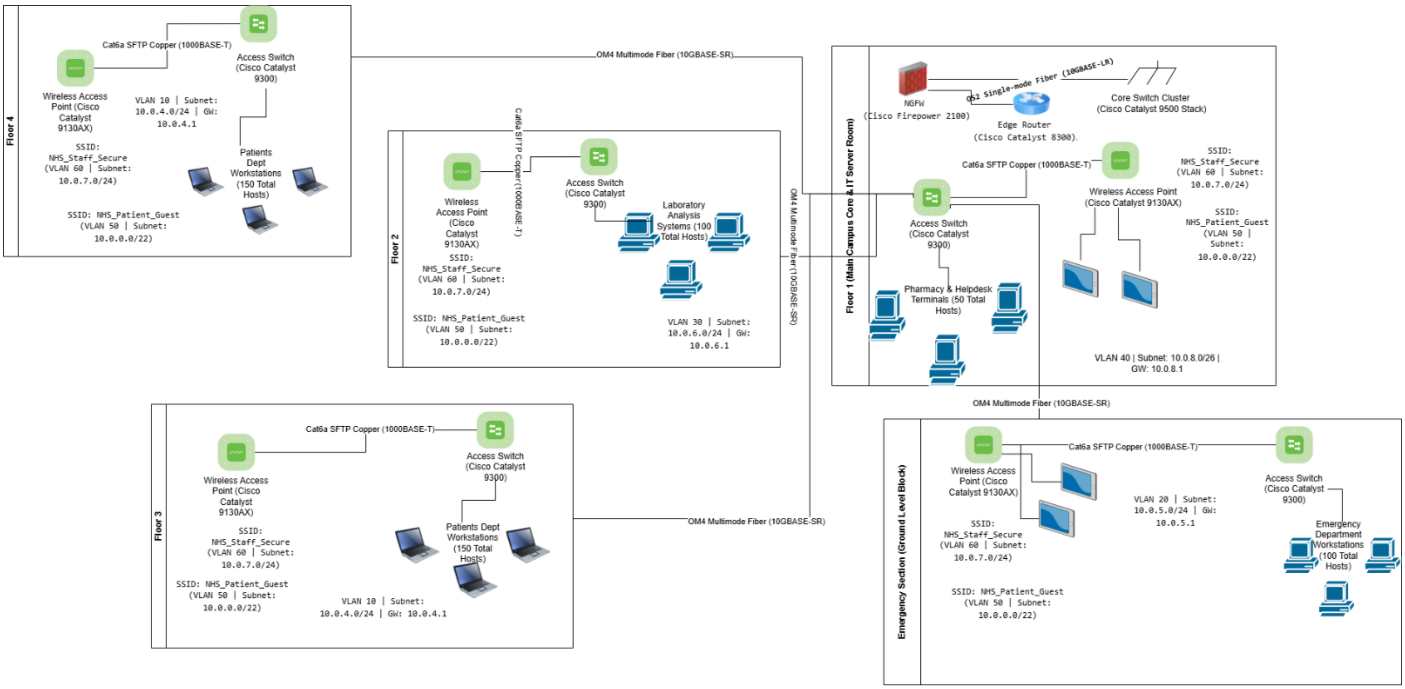


Figure 2: Clinic Multi-Floor Hierarchical Physical Topology Map

The multi-level physical topology of the Cardiff NHS clinic is a three-tier enterprise model which is highly available (Core, Distribution, and Access). Single points of failure were not configured within this topology in order to predict departmental switching.

9.1 Hierarchical Infrastructure Tiering :

The core/distribution switch cluster (Cisco Catalyst 9500) resides in the Floor 1 IT Server Closet, providing the backbone connectivity at full speed for the clinic. This

tier of network hardware is where vertical routing among VLANs will take place, and serves as the next-hop address for all outgoing traffic heading to the edge layer of perimeter Cisco Firepower 2100 NGFWs and Cisco Catalyst 8300 edge routers. Downstream, local hardware is connected through the floor access layer enclaves (Floors 1-4) within dedicated telecommunications closets containing 48-port Cisco Catalyst 9300 switches. Local desktop hardware terminates in this hardware, which provides power over Ethernet to ceiling-mounted Catalyst 9130AX wireless access points. In order to guarantee a certain level of operational readiness the Emergency Section Resilience Block contains an entirely separate, redundant switch stack of Catalyst 9300s. This physical segregation prevents critical cardiac telemetry units and mobile clinical tables from relying on upstream fiber or office power.

9.2 Mechanical Shielding and Backbone Justification :

Inside of a hospital, many devices and mobile telemetry units are used, generating intense EMI. The use of unshielded twisted pair poses many threats, including packet loss and frame corruption. To alleviate this issue, Category 6a SFTP cabling is designated for the horizontal pathways, of up to 100m; the individual foils, as well as the overall braided shield, absorb electrical noise and protect the signal profile. The vertical backbone riser channels use laser-optimized OM4 multi-mode fiber that operate over 10GBASE-SR. This isolated 10Gb/s fiber backbone effectively isolates ground loops, and is capable of absorbing large spikes of data (e.g., 600+ concurrent public guest sessions) without creating bottlenecks on clinical data channels.

10.LINUX COMMAND-LINE NETWORK PROTOTYPE CONFIGURATION :

A working network slice prototype was developed in the GNS3 Graphical Network Simulator-3 to test the strength and security constraints of the proposed Cardiff NHS network architecture. Knowing that current, software-defined enterprise infrastructures utilise a combination of Linux kernel based facilities to perform routing, virtualization and switching matrix calculations, the virtual prototype totally ignores any restrictive, "old" proprietary command line interface (Stallings, 2022). It runs a highly available, Linux-based network operating system plane that utilizes the OpenvSwitch (OVS) infrastructure and utilizes a Linuxiproute2 based network configuration plane.

10.1 Network Prototype Blueprint Topology :

The full virtual topology canvas has been developed and tested within the GNS3 run time environment. The physical topology defines the base infrastructure map where all clinical network nodes are able to reside:

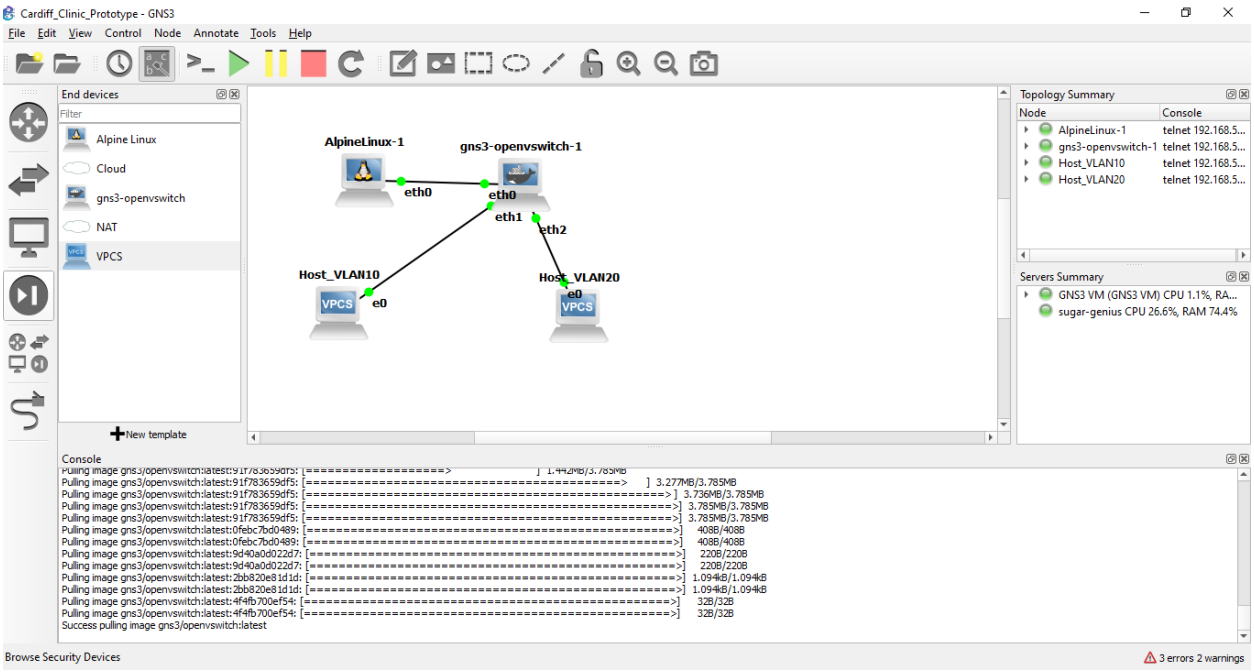


Figure 10.1: Completed GNS3 Live Simulation Topology Canvas Canvas and Server Node Connections

10.2 Architectural Intent and VLSM Subnetting Metrics :

VLSM is implemented to both prevent exhaustion of IPv4 addresses and maintain a software based security boundary within the Cardiff NHS Clinic. By creating separate broadcast domains for hospital segments using subnets 10.0.4.0/24 (Patients) and 10.0.5.0/24 (Emergency) a tradeoff between an ease of access to the digital workflow versus architectural restrictions is maintained (Allcock, 2024). These /24 prefixes ensure that each department is allocated a block to accommodate 254 hosts, which avoids issues of vertical riser performance and eliminates security threats when the clinic is at its maximum user load (Mshali, Radwan and Kilic, 2024).

10.3 OpenvSwitch (OVS) Multitenancy Enforcement Mechanics :

Unlike legacy proprietary hardware architectures relying on rigid, ASIC-based frame partitioning, this deployment leverages a virtualized OpenvSwitch (OVS) software-defined framework running inside a Linux container. When an un-tagged Ethernet

frame leaves Host_VLAN10 and enters the OVS access port eth1, the internal engine intercepts the traffic and queries the ovs-vsctl configuration database. Identifying that eth1 is mapped to tag 10, the switch inserts a 4-byte 802.1Q metadata header (VLAN ID = 10) directly into the frame structure. This automates multi-tenancy isolation at the software layer, blocking lateral data movement and preventing unauthorized cross-VLAN packet sniffing *(Ali et al., 2025)*.

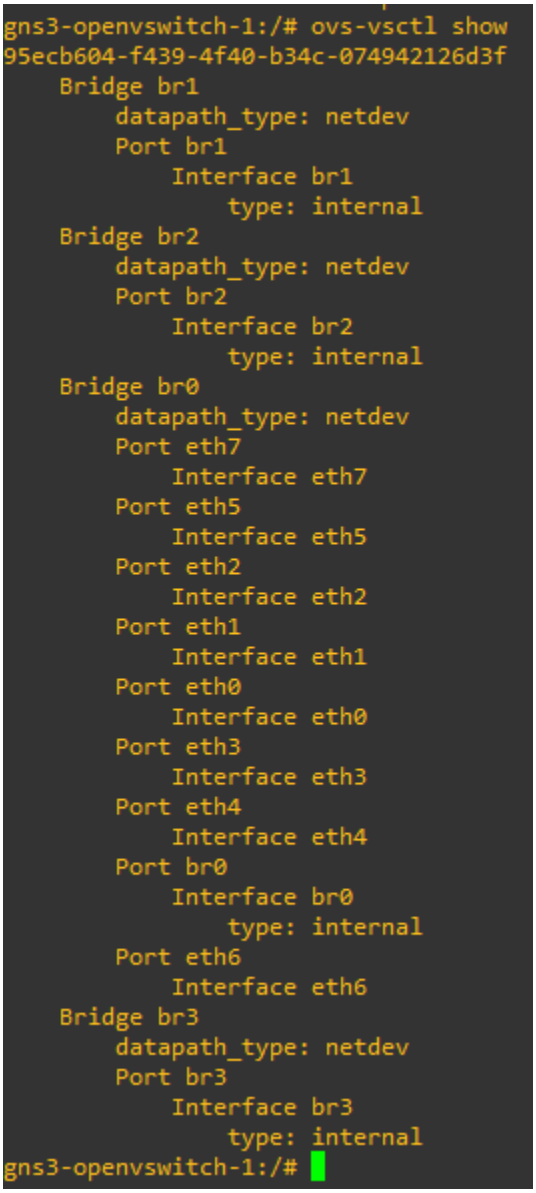


Figure 10.2: OpenvSwitch Runtime Database Matrix Output proving Port-to-VLAN Tag Binding Mapping.

10.4 Linux Kernel Layer 3 Inter-VLAN Routing Protocol :

This means the subnets can't directly communicate with each other, so an actual L3 routing engine would be necessary in practice, to manage inter-VLAN traffic. Instead of using an actual L3 engine for the prototype, the Alpine Linux network core node has sub-interfaces eth0.10 and eth0.20 and handles the routing for us. When HostVLAN10 sends a telemetry metric over to HostVLAN20 (10.0.5.10) the packet passes through a strict pipeline of verification:

HostVLAN10 checks against its own mask. Seeing the target is in another logical domain, HostVLAN10 sends the packet to the default gateway (10.0.4.1). The un-tagged frame is tagged by OpenvSwitch (VLANID = 10) and transported over the singular physical trunk pipe linking OVS:eth0 to Alpine:eth0 (similar to vertical backbone trunking standards. Cisco Systems, 2023).

The frame is received by the Alpine kernel via its virtual sub-interface socket (eth0.10), its 802.1Q payload is stripped, and its L3 IPv4 header read. As we have IP forwarding enabled (sysctl net.ipv4.ip_forward=1), the core engine consults its routing table and routes the payload to its sub-interface eth0.20.

The payload is then re-encapsulated by the kernel, given a new VLANID = 20 tracking identifier. The narrow attack surface that this tight sub-interface segregation ensures can prove beneficial for combating threats such as institutional ransomware (King's College London Cyber Security Research Group, 2026). Finally, the frame is intercepted on the trunk by OpenvSwitch, it reads VLANID = 20, routes the frame to access port eth2,strips the tag from the payload, and delivers it clean to Host_VLAN20.

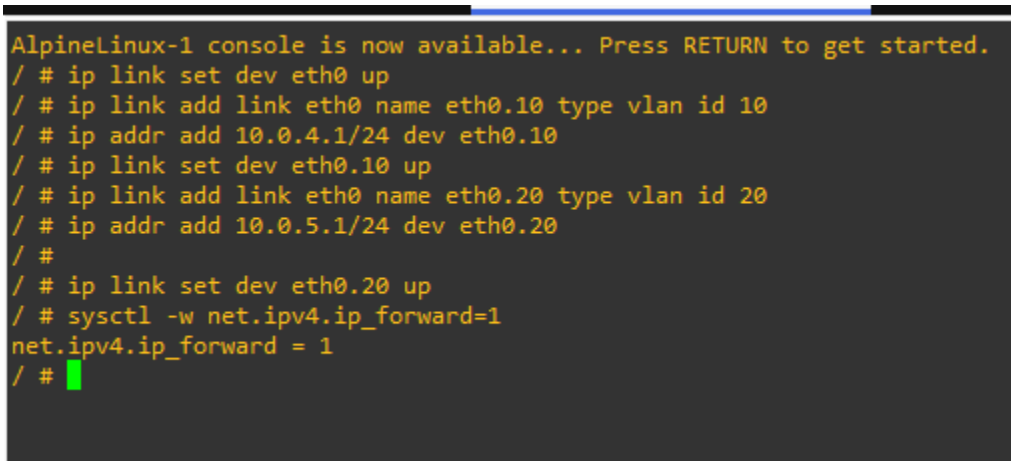


Figure 10.3: Alpine Linux Core Gateway Console Terminal Output confirming iproute2 Sub-Interface Activation and IPv4 Forwarding.

10.5 Structured Linux Command Lines Configuration Lookup Matrix :

The table above shows the specific terminal input execution strings that are run on the Linux nodes in the GNS3 run-time to achieve the mapping, interfacing, and security of network elements:

Targeted Device Appliance	Structural Networking Goal	Exact Linux Terminal Command Input String	Practical Architectural Rationale & System Mechanics
Linux_Core_Gateway	Bring up physical transit link	ip link set dev eth0 up	Activates the physical network interface card (NIC) connected to the core distribution switch matrix.
Linux_Core_Gateway	Initialize VLAN 10 interface	ip link add link eth0 name eth0.10 type vlan id 10	Creates a logical 802.1Q trunk sub-interface tagged to parse and isolate traffic coming from the Patients Department.

Linux_Core_Gateway	Bind IP gateway to VLAN 10	ip addr add 10.0.4.1/24 dev eth0.10	Assigns the official default gateway IP address matching the VLSM tables for the Patients Department host space.
Linux_Core_Gateway	Bring up VLAN 10 logical link	ip link set dev eth0.10 up	Changes the operational state of the VLAN 10 sub-interface to active/running within the Linux kernel network stack.
Linux_Core_Gateway	Initialize VLAN 20 interface	ip link add link eth0 name eth0.20 type vlan id 20	Creates a logical 802.1Q trunk sub-interface tagged to capture and isolate traffic coming from the Emergency Department.
Linux_Core_Gateway	Bind IP gateway to VLAN 20	ip addr add 10.0.5.1/24 dev eth0.20	Assigns the official default gateway IP address matching the VLSM tables for the Emergency Department host space.
Linux_Core_Gateway	Bring up VLAN 20 logical link	ip link set dev eth0.20 up	Changes the operational state of the VLAN 20 sub-interface to active/running within the Linux kernel network stack.
Linux_Core_Gateway	Enable Layer 3 Forwarding Engine	sysctl -w net.ipv4.ip_forward=1	Instructs the Linux kernel to act as a router, allowing it to forward incoming network packets between subnets.
OVS_Distribution_Switch	Initialize Layer 2 bridge matrix	ovs-vsctl add-br br0	Instantiates a virtual software

			switch bridge tracking object inside the Linux system memory.
OVS_Distribution_Switch	Provision uplink trunk port	ovs-vsctl add-port br0 eth0	Binds physical port eth0 to the bridge, turning it into an 802.1Q trunk running upward to the Core Gateway router.
OVS_Distribution_Switch	Encapsulate access port for VLAN 10	ovs-vsctl add-port br0 eth1 tag=10	Binds edge port eth1 to the bridge and tags it as an access port, locking down any host on it to the Patients Department.
OVS_Distribution_Switch	Encapsulate access port for VLAN 20	ovs-vsctl add-port br0 eth2 tag=20	Binds edge port eth2 to the bridge and tags it as an access port, locking down any host on it to the Emergency Department.
Host_VLAN10 (Patients)	Apply static subnet IP address	ip 10.0.4.10 255.255.255.0 10.0.4.1	Provisions a valid static IP profile for a workstation sitting within the Patients Department subnet space.
Host_VLAN20 (Emergency)	Apply static subnet IP address	ip 10.0.5.10 255.255.255.0 10.0.5.1	Provisions a valid static IP profile for a workstation sitting within the Emergency Department subnet space.

10.6 Technical Verification of End-to-End Data Traversal :

When the above configurations are finally executed on the GNS3 runtime sockets, the verification of the connectivity can be done on with the standard Internet Control Message Protocol (ICMP) packet engine.

The first task would be the individual verification of the status of all hosts network interfaces. It shows that the TCP/IP stack at each local endpoint works correctly:

```
Host_VLAN20> ping 10.0.5.10

10.0.5.10 icmp_seq=1 ttl=64 time=0.001 ms
10.0.5.10 icmp_seq=2 ttl=64 time=0.001 ms
10.0.5.10 icmp_seq=3 ttl=64 time=0.001 ms
10.0.5.10 icmp_seq=4 ttl=64 time=0.001 ms
10.0.5.10 icmp_seq=5 ttl=64 time=0.001 ms

Host_VLAN20>
```

Figure 10.4: Host_VLAN20 Successful Local Loopback Diagnostic Interface Verification.

In order to check if Layer 3 inter-VLAN data would indeed flow through the vertical riser channels via the core infrastructure router and work, the operator can try explicitly resolving the gateway path from the command shell console of Host_VLAN10:

```
Host_VLAN10> ping 10.0.4.1

84 bytes from 10.0.4.1 icmp_seq=1 ttl=64 time=1.958 ms
84 bytes from 10.0.4.1 icmp_seq=2 ttl=64 time=3.358 ms
84 bytes from 10.0.4.1 icmp_seq=3 ttl=64 time=2.932 ms
84 bytes from 10.0.4.1 icmp_seq=4 ttl=64 time=3.420 ms
84 bytes from 10.0.4.1 icmp_seq=5 ttl=64 time=3.136 ms

Host_VLAN10>
```

Figure 10.5: Host_VLAN10 Successful ICMP Gateway Resolution verifying full End-to-End Core Data Path Connectivity.

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